A Hardware Rendering of the Rho Calculus with Boolean Arithmetic and Builtin Data

Design specification: source language, target machine, and verification guidance

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Abstract

This note specifies a hardware rendering of the rho calculus extended with Boolean arithmetic and a set of builtin data types, as the target of a compiler from f1r3lang. It fixes the source language (syntax, sorts, structural congruence, matching modulo the Boolean laws, and reduction), the canonical forms that make names comparable in constant time, and a machine that tiles the surface of an FPGA with small processors. Each processor owns a namespace given by a structural formula, reduces in the same clock cycle as every other, and exchanges processes with its neighbours over a mesh. The note also gives the organisation of the implementation in Bluespec and the obligations an implementation must discharge to count as verified. Decisions marked C are confirmed; those marked P are proposed by this draft. Draft 0.5 adds the builtin data types; the changes of this and earlier drafts are listed in Section 12.

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1 Purpose and scope

The goal is a machine whose instruction set *is* the calculus. Three things follow from taking that literally.

- The only state change is the COMM rule.
- Every source of nondeterminism in the calculus is a physical arbitration in the machine.
- Every independence in the calculus is physical concurrency in the machine. Reductions on channels in different namespaces happen in the same clock cycle, on different processors, with no scheduler between them.

The machine is therefore not one processor but a surface of them: the FPGA is tiled with identical processors, each assigned a namespace, and the running process is spread over the tiles according to the channels its components use.

The first target is an FPGA, described in Bluespec, which emits Verilog. The design is restricted to components expressible in synthesizable Verilog or VHDL for CMOS. Clockless and superconducting realisations are out of scope, but every interface between and within tiles is latency-insensitive, so a later self-timed version does not require a redesign.

The machine is meant as the target of a compiler from f1r3lang, which is this calculus with graded `where` clauses and data. The calculus here is the kernel. The data types are those of Section 5; the `where` clause is the next addition, and its values and the choice it drives are already provided for.

In scope: the calculus of Section 3; the builtin data types of Section 5; the tile, the array and the mesh; a compiler from source text to a memory image; a verification plan. Out of scope for this draft: cost accounting, join patterns, `where` clauses and grading, the combinator instruction set, and arrays spanning more than one device. Hooks are left where these attach (Sections 6.9 and 10).

2 Decision register

ID	Decision	Status
D1	Boolean operands admit drops: $*x$ is a Boolean expression.	C
D2	The Boolean-algebra laws are part of structural congruence.	C
D3	Replication is derivable by the reflective encoding; a persistent receive is provided as sugar and implemented as a primitive.	C
D4	Ill-sorted terms are excluded by a basic spatial-behavioural type system, not by the machine.	C
D5	Distinct solutions of a Boolean pattern are distinct, observable races, on the same footing as COMM races.	C
D6	First target is an FPGA, via generated Verilog.	C
D7	A high-level functional hardware language is used in place of hand-written RTL.	C
D8	Each processor is a tuple-space machine: hash-consed term heap, send and receive tables, matcher, arbiter. Static elaboration of the Boolean-typed fragment is a later optimisation.	C
D9	The whole design is written in Bluespec (BH syntax). Clash versions of the pure datapath functions may follow. One Haskell reference library is shared with the compiler.	C
D10	The compiler is trusted for sorts. The machine keeps a sticky `FAULT_SORT` flag whose unreachability is a proof obligation. The flag is a debugging aid: it does not detect every ill-sorted program.	C
D11	Variables are a separate syntactic category V, instantiated in the machine as de Bruijn indices. Names remain quoted processes.	C
D12	Open Boolean code is canonicalised as a truth table over at most $K = 6$ variables; larger supports are rejected at compile time.	C
D13	Boolean patterns (other than a bare variable) match closed Boolean values only.	C
D14	Fairness is possibilistic: every enabled redex has nonzero probability at every step. Uniformity over redexes is not required.	C
D15	Entropy is an LFSR behind an interface; every arbitration is written to a decision log that makes runs replayable.	C
D16	The machine is board-agnostic: no vendor primitives, inferred block RAM only, all sizes are parameters.	C

ID	Decision	Status
D17	The machine is an array of identical processors (tiles) covering the FPGA, joined by a nearest-neighbour mesh.	C
D18	Each tile owns a namespace. Namespaces are given by structural formulae on the quoted process.	C
D19	A tile serves several channels and commits at most one COMM per clock cycle. The default array has about 100 tiles, each with about one hundredth of the device's memory.	C
D20	Reduction is concurrent. There is no global scheduler and no sequential reactor: tiles act in the same cycle, and inside a tile the stages are concurrent rules joined by queues.	C
D21	Namespace formulae are drawn from the fragment recognised by a deterministic bottom-up tree automaton. A name's class is computed when it is interned, and ownership is a table lookup on the class.	P
D22	Heaps are per tile. Only closed terms cross between tiles, as serialised packets that are re-interned on arrival. Static code is replicated in every tile.	P
D23	Routing is dimension-ordered with wormhole packets. A tile's inbound path never waits on its outbound path.	P
D24	Every clock edge is a boundary. Each tile keeps its own decision log, stamped with the cycle; the array's log is a partial order.	P
D25	Builtin data comes in two kinds: inline types evaluated in every tile, and service units reached as channels in their own namespaces.	P
D26	A type is builtin only if it has a canonical form, bounded total operations, exact determinism, and a hopeless encoding.	P
D27	`Int` is 64-bit, signed and wrapping. `Bytes`, strings, tuples, lists, sets and maps are inline; hashing, signature checking and vector arithmetic are service units.	P
D28	Grades are fixed-point or integer, chosen per resolution algebra, so that sums over candidates are exact and independent of order.	P
D29	Outside the Booleans, congruence contains evaluation and the laws of collections only. There is no arithmetic in patterns.	P
D30	A partial operation with no result raises `FAULT_DATA`.	P
D31	Leaf data may be classed by sort and by a residue of its content, so that channels named by strings or numbers spread over tiles.	P
D32	Floating point is deferred. If added it is a service unit for data, never a grade, under the four conditions of Section 5.4.	P

3 Source language

3.1 Syntax

Let V be a countable set of variables, ranged over by v, w. Processes P, Q, Boolean expressions B, names x, y, patterns p and deep patterns d are given by

$$
\begin{array}{rcl}
P, Q & ::= & \mathtt{0} \mid B \mid \mathtt{for}(p \leftarrow x)\, P \mid \mathtt{for}(p \Leftarrow x)\, P \mid x!(Q) \mid P \mid Q \mid *x \\
B & ::= & \mathtt{true} \mid \mathtt{false} \mid *x \mid \sim B \mid B \mathbin{\&\&} B \mid B \mid\mid B \mid B \mathbin{\mathtt{xor}} B \\
x, y & ::= & @P \mid v \\
p & ::= & v \mid @d \\
d & ::= & \mathtt{0} \mid \mathtt{true} \mid \mathtt{false} \mid v \mid p!(d) \mid \mathtt{for}(v \leftarrow p)\, d \mid \sim d \mid d \mathbin{\&\&} d \mid d \mid\mid d \mid d \mathbin{\mathtt{xor}} d
\end{array}
$$

Relative to the grammar that opened the session, **D1** adds $*x$ to B, **D3** adds the persistent receive $\mathtt{for}(p \Leftarrow x)\, P$, and **D11** separates variables from names. Quotation is written $@P$ and

dereference $*x$ throughout. The production $*x$ is shared by P and B; which reading applies is settled by sorts.

A receive binds the variables of its pattern in its body. Every variable denotes a *name*. A variable in a deep-pattern position denotes the name of the matched subprocess, so matching $@(v \mathbin{\&\&} w)$ binds v and w to names of Booleans, and the body uses them as $*v$ and $*w$. A program is a closed process.

3.2 Sorts

There are two sorts of process, $\mathsf{Bool} \subseteq \mathsf{Proc}$, and correspondingly two sorts of variable: *Boolean-sorted* variables, which only ever denote $@\texttt{true}$ or $@\texttt{false}$, and *structural* variables, which may denote any name. A term is well sorted when every operand of a Boolean operator is of sort Bool; in particular $*v$ as an operand requires v to be Boolean-sorted, and a Boolean-sorted variable requires every message that can reach its receive to quote a Boolean.

The last condition is a property of channels, and is discharged by the spatial-behavioural type system (**D4**), which assigns a carried sort to each namespace. This note treats that system as a black box with the following interface.

- **Input.** A closed process.
- **Output.** Either a rejection, or a sort for every pattern variable, such that reduction preserves well-sortedness.

The reflective encoding of replication sends a quoted process on a channel that the process itself mentions, so the system needs a recursive formula to type it. Giving the persistent receive its own typing rule avoids that for programs that use only the sugar.

3.3 Structural congruence

Structural congruence $\equiv$ is the least congruence on processes containing:

1. alpha-equivalence of bound variables;
2. $(P \mid Q) \mid R \equiv P \mid (Q \mid R), \quad P \mid Q \equiv Q \mid P, \quad P \mid \mathtt{0} \equiv P$;
3. the laws of Boolean algebra on B, with $\texttt{xor}$ defined by $B_1 \mathbin{\texttt{xor}} B_2 \equiv (B_1 \mathbin{\&\&} {\sim}B_2) \mid\mid ({\sim}B_1 \mathbin{\&\&} B_2)$;
4. $*@P \equiv P$.

Name equivalence $\equiv_N$ is the least equivalence with $@P \equiv_N @Q$ whenever $P \equiv Q$, and $@{*x} \equiv_N x$. Law 4 is adopted as a convention; in the original presentation the same effect arises only through substitution.

By **D2** every closed Boolean expression is congruent to exactly one of $\texttt{true}$ or $\texttt{false}$, so there are exactly two closed Boolean names up to $\equiv_N$. The requirement that substituting a Boolean name into $*y$ yields the *evaluation* of the Boolean is therefore a consequence of law 3 and not a separate rule: evaluation is normalisation.

3.4 Matching

A substitution σ maps variables to closed names and respects sorts.

Definition 1 (Match set)**.** For a pattern p and a closed name n,

$$\mathsf{match}(p, n) \;=\; \{\, \sigma \mid \mathrm{dom}(\sigma) = \mathrm{vars}(p),\ p\sigma \equiv_N n \,\}/{\equiv_N}$$

where substitutions are identified when they agree pointwise up to $\equiv_N$.

Matching is *modulo* $\equiv$. For structural patterns this is ordinary tree matching on normal forms, and the match set has at most one element. For Boolean patterns it is equation solving: $\mathsf{match}(@(v \mathbin{\&\&} w), @\mathtt{false})$ has three elements and $\mathsf{match}(@(v \mathbin{\&\&} w), @\mathtt{true})$ has one. Because Boolean-sorted variables range over two classes, a pattern with k Boolean variables has at most 2^k solutions.

Restrictions.

- (**D13**) A Boolean pattern built with an operator matches only a closed Boolean value. Matching an operator pattern against open Boolean code under a binder would be unification of Boolean functions and is excluded. A Boolean leaf, a constant leaf included, that meets a name not quoting a Boolean is a sort fault and not a failed match; so the pattern $@\mathtt{true}$ cannot be used to test whether a message is a Boolean.
- A pattern variable is a *mask variable* when it occurs under a Boolean operator in the pattern; every occurrence of it is then a Boolean leaf. Any other variable is bound structurally, whatever its sort, and its sort is checked where the body uses it. A pattern has at most K mask variables (**D12**).
- Deep patterns contain no parallel composition, as in the given grammar, so no matching modulo associativity and commutativity is needed.
- The receive pattern $\mathtt{for}(v \leftarrow p)\, d$, which matches under a binder, is deferred; see Section 11.

3.5 Reduction

$$\frac{x \equiv_N y \qquad \sigma \in \mathsf{match}(p, @Q)}{\mathtt{for}(p \leftarrow x)\, P \mid y!(Q) \;\rightarrow\; P\sigma} \;(\textsc{comm})$$

$$\frac{x \equiv_N y \qquad \sigma \in \mathsf{match}(p, @Q)}{\mathtt{for}(p \Leftarrow x)\, P \mid y!(Q) \;\rightarrow\; P\sigma \mid \mathtt{for}(p \Leftarrow x)\, P} \;(\textsc{comm}!)$$

$$\frac{P \rightarrow P'}{P \mid Q \rightarrow P' \mid Q} \;(\textsc{par}) \qquad \frac{P \equiv P' \quad P' \rightarrow Q' \quad Q' \equiv Q}{P \rightarrow Q} \;(\textsc{equiv})$$

There is no reduction under a receive or under a quote. Substitution $P\sigma$ replaces free variables by names and does not descend into closed quoted terms, which have no free variables.

A *redex* of a process is a triple (send, receive, solution). By **D5** two redexes that differ only in the solution are different transitions. A send whose channel has a receive but whose payload admits no solution is not a redex and stays in place.

3.6 Replication

With $D(x) = \mathtt{for}(v \leftarrow x)\,(x!(*v) \mid *v)$, replication of P on a name x not otherwise used is $x!(D(x) \mid P) \mid D(x)$. The machine must run this term correctly without special support, which requires a general drop of quoted code. The persistent receive is the efficient form. Its specification is the encoding:

Obligation 1. For all p, x, P, the process $\mathtt{for}(p \Leftarrow x)\, P$ is weakly bisimilar to its reflective encoding, under context-labelled transitions.

4 Canonical forms

The machine compares names by comparing machine words. That is sound only if every term is stored in a normal form that is unique up to $\equiv$. The normal form $\mathsf{nf}(P)$ is defined bottom-up.

Variables. Bound variables are replaced by de Bruijn indices: an occurrence is the number of variable binders between it and its own binder, counting each variable of a multi-variable pattern in left-to-right order of first occurrence. Closed subterms therefore have a representation that does not depend on their context, which is what permits sharing. This indexes V only; names are still quoted processes (**D11**).

Quote and drop. $*@P$ is replaced by $\mathsf{nf}(P)$, and $@*x$ by x.

Booleans. A Boolean expression whose operands are all closed is replaced by its value. Otherwise let $v_0 < \cdots < v_{k-1}$ be the variables it *depends on* (its support), ordered by index. The normal form is the pair of the support and the truth table $T \in \{0,1\}^{2^k}$, where bit $r = \sum_i b_i 2^i$ of T is the value of the expression at $v_i \mapsto b_i$. Variables on which the function does not depend are removed. A function that is the identity on a single variable v is stored as $*v$. If $k > K$ the program is rejected (**D12**).

Parallel composition. Compositions are flattened, `0` units are removed, and the components are sorted into ascending order of their machine words. Since each component is itself canonical and hash-consed (Section 6.5), word order is a total order on components up to $\equiv$. Duplicates are kept: composition is a multiset.

Other constructors. Sends, receives and persistent receives are normalised componentwise. Patterns are normalised by the compiler into pattern programs (Section 8); two receives are congruent only if their pattern programs are identical.

Obligation 2 (Canonicity)**.** For all well-sorted P, Q within the size bounds: $\mathsf{nf}(P) = \mathsf{nf}(Q)$ if and only if $P \equiv Q$.

Remark 1. The truth table over a minimal, ordered support is a canonical form for Boolean functions, so the Boolean part of Obligation 2 is immediate. The same 64-bit word serves three purposes: it is the normal form, it evaluates the function forwards (index it by the argument values), and it solves patterns backwards (Section 6.8).

5 Builtin data types

The calculus of Section 3 is the kernel of f1r3lang, and Booleans are its only data. A compiler from f1r3lang needs more: rholang's ground types and collections, and the values of graded `where` clauses. All of these can be encoded as processes, and none should be. This section says which types the machine provides directly, and by what rule.

5.1 The admission test

A type is builtin when all four of these hold (**D26**).

1. **Canonical form.** Equal values have equal representations, so that a name quoting a value is compared in one word, like any other name.
2. **Bounded, total operations.** Each operation is a function that finishes in a time bounded by the size of its operands.
3. **Exact determinism.** The result of an operation is the same in every tile, on every device, and in the reference. Nothing is implementation-defined.
4. **A hopeless encoding.** The encoding in the kernel costs orders of magnitude more reductions than the hardware operation costs gates.

The Booleans pass, and are the model. A 32-bit addition encoded as Boolean processes is hundreds of COMMs, each a tile cycle and often a trip across the mesh; as a builtin it is evaluated inside the one COMM that needed it.

5.2 Two kinds of builtin

Inline types are evaluated in every tile, at instantiation, as Booleans are. Their operations are small enough to sit in the datapath of each of a hundred tiles: integer and fixed-point arithmetic, comparison, and the constructors of collections.

Service units are too large to replicate: hash functions, signature checks, wide multiply-accumulate. A service unit is a tile of a different kind that owns a namespace of *service channels*. A request is an ordinary send on a service channel, carrying its arguments and a return name; the reply is an ordinary send on the return name. Nothing in the calculus or in the other tiles changes, and a program cannot tell a service unit from a process that computes the same function.

The second kind is how rholang already presents its system contracts, and the namespace map of Section 6.2 is what places them (**D25**).

5.3 The types

Type	Kind	Values and operations
`Bool`	inline	as in Sections 3 and 4
`Int`	inline	64-bit two's complement, wrapping. `+ - *`, `/ %` truncating, comparisons, bitwise operations and shifts
`Fix`	inline	fixed-point numbers: a width of at most 64 bits and a number of fraction bits, signed or unsigned. Exact `+ -`; `*` truncated towards $-\infty$; comparisons; `min`, `max`
`Bytes`	inline	immutable byte strings. Length, concatenation, slice, index, conversion to and from `Int`
`String`, `Uri`	inline	`Bytes` holding UTF-8, under a different sort. A `Uri` has no operations but equality
tuple	inline	fixed arity; projection
list	inline	head, tail, length, index, concatenation
set	inline	union, difference, membership, insertion, deletion, size
map	inline	lookup, update, deletion, membership, keys, size
hash	service	BLAKE2b-256, Keccak-256, SHA-256 on `Bytes`
signature	service	secp256k1 and Ed25519 verification
vector	service	vectors of `Fix`; dot product and scaled addition

`Int` is rholang's integer. The collections contain processes, as rholang's do, so they may hold code and names as well as data.

Not builtin. Unbounded integers and rationals fail the second test and are left to processes over `Int`. A type defined in f1r3lang's data definition sublanguage is builtin only if its equations have a cheap normal form; otherwise its terms are heap nodes and its rewrites are processes.

5.4 Grades

A graded `where` clause is valued in a commutative semiring and is evaluated once per candidate match. The arbiter must then sum over a *multiset* of candidates and choose among them. The sum has to be independent of the order in which candidates are visited, or two tiles, or a run and its replay, could resolve the same race differently. That is a requirement of exact associativity and commutativity, and it decides the representation (**D28**).

Resolution algebra	Representation	Notes
Boolean $(\vee, \wedge)$	`Bool`	the kernel; choice is uniform over the solutions
Tropical $(\min, +)$	unsigned `Int`, top value for ∞	exact; addition saturates at ∞, which is associative on unsigned values
Viterbi $(\max, \times)$ on $[0, 1]$	tropical, on $-\log$	exact; literals are converted by the compiler
$\mathbb{R}_{\geq 0}$ $(+, \times)$	unsigned `Fix`	sums exact in a widened accumulator; range is limited by the width
PLN power on $[0, 1]$	unsigned `Fix`, one integer bit	as above
$\mathbb{C}$ $(+, \times)$	a pair of signed `Fix`	the type only; the interfering resolver is not in this draft

Products in a clause are indexed by positions in the program text and are taken in that order, so truncation in `*` is deterministic. Sums are indexed by candidates and are exact. The `where` clause itself remains outside this draft; what is fixed here is the type of its values and the arbiter's weighted choice (Section 6.9), so that adding the clause does not change the tile.

Floating point (D32). Floating point is not a grade and is not in this draft. Its addition is not associative, so a sum over candidates would depend on their order; it has unequal equal values (NaN) and equal unequal ones (the two zeros), which break canonical form; and on an FPGA its adder costs several pipeline stages where the hard multipliers are integer. The one thing it offers that `Fix` does not is dynamic range. When that is needed for data, floating point can be added as a service unit under four conditions: one rounding mode, one canonical NaN, one zero, and sums taken in the canonical order of their operands.

5.5 Laws

The Boolean laws are in structural congruence because a truth table is a cheap canonical form for an open Boolean expression. Arithmetic has none, so for every other type congruence contains only two things (**D29**).

- **Evaluation.** A closed application of an operation to values is congruent to its result. An open application is syntax: it is stored as written and evaluated when instantiation closes it.
- **Collections.** A set is unordered and without duplicates; a map has one entry per key. Their normal forms are sorted, as parallel composition is.

The laws of rings are not in congruence: $*x + 1$ and $1 + *x$ are different open terms that become equal when x is known.

An operation with no result (division by zero, an index out of range) raises `FAULT_DATA`. A sort error in an operand always raises `FAULT_SORT`, because no law can erase the operand first; the gap noted in Section 6.13 is peculiar to the Booleans.

5.6 Patterns

A pattern may contain a literal of any inline type, a tuple of patterns, and a list pattern giving a head and a tail. It may not contain an arithmetic operation: where a Boolean operator pattern has at most 2^K solutions, $p + q$ against a 64-bit value has 2^{64}, and a filter on values is the business of the `where` clause. Sets and maps cannot be taken apart by patterns, since that is matching modulo commutativity; they are examined by their operations.

6 Target machine

6.1 Overview

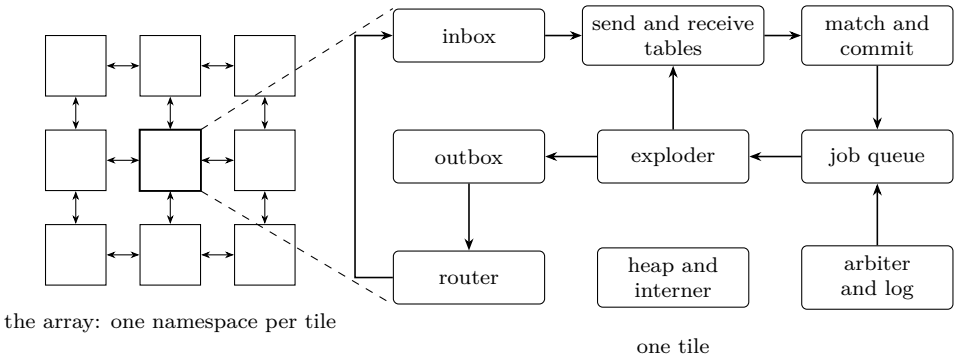


Figure 1: The array and one tile. Inside the tile, arrows show the way processes flow; each box acts in every cycle in which its inputs are ready.

The machine is a $W \times H$ array of identical *tiles*. A tile is a complete processor for the calculus, restricted to the channels of its own namespace. The running process is spread over the array as a *soup*: each top-level send or receive rests in the tile that owns its channel, or is in transit towards it. Everything quoted lives in the heap of the tile that holds it, as a hash-consed canonical term.

A COMM is a local event in one tile: the tile selects a redex among the sends and receives resting in it, removes them, and queues the instantiated continuation. The continuation is then taken apart, and each of its components is delivered to the tile that owns its channel, which may be the same tile or a distant one.

Tiles do not take turns. In one clock cycle every tile that holds a redex may commit one, and no tile needs to know what any other is doing, because redexes in different tiles share no send and no receive. Rho sends carry no continuation, so the time a component spends in transit is not observable, and the calculus needs no change to be distributed in this way.

6.2 Namespaces and ownership

Structural formulae. A namespace is a set of names, given by a formula on the process a name quotes. Formulae φ and name formulae α are

$$\begin{aligned} \varphi &::= \top \mid \mathtt{0} \mid \mathtt{true} \mid \mathtt{false} \mid \alpha!(\varphi) \mid \mathtt{for}(\alpha)\,\varphi \mid \Diamond\varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid \neg\varphi \\ \alpha &::= @\varphi \end{aligned}$$

They are read on closed normal forms. `0`, `true` and `false` hold of exactly those processes. $\alpha!(\varphi)$ holds of a send whose channel is in α and whose payload satisfies φ. $\mathtt{for}(\alpha)\,\varphi$ holds of a receive, of either kind, on a channel in α whose body satisfies φ. $\Diamond\varphi$ holds when some parallel component satisfies φ; a process that is not a composition is its own single component. A term with a free variable satisfies no atom, so a formula sees into a receive's body only as far as the body is closed. There are no behavioural modalities and no counting of components: the formulae are structural, as **D18** requires, and satisfaction is invariant under $\equiv$ because it is defined on normal forms.

The namespace map. A map for an array of T tiles and P ports is a list of name formulae $\alpha_0, \ldots, \alpha_{T+P-1}$ that *partition* the closed names: every closed name satisfies exactly one. The tile or port t *owns* the names that satisfy α_t. The compiler checks the partition and rejects a map that fails it.

Realisation (D21). The compiler turns the map into one deterministic bottom-up tree automaton. Its states are called *classes*; there are at most 2^c of them.

- The class of a node is a function of its kind and the classes of its fields, given by small tables: $\delta_{send}(c_{chan}, c_{payload})$, $\delta_{recv}(c_{chan}, c_{body})$ and $\delta_{par}(c_{head}, c_{tail})$. Immediates have fixed classes, and so does every open term.
- δ_{par} is associative and commutative. It depends only on the set of component classes and on whether there is more than one component, which is what $\Diamond$ and the other atoms can observe.
- The class is computed when the node is interned and kept in the node header. The class of a name is the class of the process it quotes.
- A table τ of 2^c entries maps a class to its owner. Ownership of any name is therefore one lookup, from the name word and the local heap alone.
- The tables are identical in every tile, so a term has the same class wherever it is interned.

Partition is decidable for this fragment: disjointness is emptiness of an intersection and exhaustiveness is universality of a union, both on tree automata.

The default map. Symbols are numerals (Section 8), and "the numeral k with $k \equiv t \pmod{T}$" is regular: the class counts the depth of the numeral modulo T. The default map sends the numeral k to tile $k \bmod T$ and every other name to one designated tile. Under it, the compiler controls placement entirely by choosing which numeral each symbol receives. A program that mints channel names reflectively should supply its own map.

Data in names (D31). Formulae gain an atom for each sort of Section 5, and for leaf data an atom $leaf_{k/n}$ that holds when a fixed hash of the value is congruent to k modulo n. The second is what spreads channels named by strings or numbers, as rholang programs name them, over the array; without it they would all fall in the tile that owns their sort.

What ownership buys.

- **Races are local.** Two redexes compete only if they share a send or a receive, hence a channel, hence a tile. All arbitration is inside one tile.
- **Independence is physical.** Redexes in different tiles share nothing and commute, so they may fire in the same cycle.
- **Distance has a shape.** If the map comes from a tree of formulae, each tile's formula a conjunction along a path, then laying the tree out as an H-tree makes distance in the tree track distance on the mesh.

6.3 The array and the mesh

Geometry. Tiles are addressed by coordinates (x, y). Each has links to its four neighbours; a link is a pair of one-way channels with a valid/ready handshake, one flit wide. Ports sit beyond the edge of the mesh and are addressed as if they were tiles.

Packets (D22). A top-level send or receive travels as one packet.

- The head flit carries the destination, the kind (send or receive) and the length.
- The body is a *term stream*: the nodes of the term in post-order. Each field of a node is an immediate, a reference to an earlier node of the same packet, or a static address.

- The destination interns the nodes in arrival order. The stream is an interning script, and it preserves the sharing of the term.
- Only closed terms are sent. A receive closure bound for another tile is instantiated first and travels as a closed receive, arriving with an empty environment.

The static segment. The program's code, as compiled, is loaded at the same addresses in every tile's heap. It is read-only and never freed. A packet names a static node by its address, so program code, which is most of what moves, costs one flit however large it is.

Routing (D23). Routing is dimension-ordered: a packet travels along x to the destination column, then along y. It is wormhole switched: a packet holds each link from its head flit to its tail flit, and packets do not interleave on a link.

No deadlock. Dimension-ordered routing on a mesh cannot deadlock provided every destination consumes what arrives. The tile guarantees that: its inbound path (link, inbox, interner, tables) depends on no outbound resource. When the heap or a table is full the tile raises a resource fault, which halts the array; it never refuses a packet. Backpressure exists only on the forward path from commit to outbox, where a commit simply waits while the job queue is full.

Memory (D19). A device with M bits of block RAM and T tiles gives each tile M/T. The reference split of a tile's share is 60% heap nodes, 15% hash-cons table, 10% reference counts and class headers, 10% tables and environments, and 5% queues. The split is a parameter.

6.4 Words and nodes

All widths are parameters; the reference values are for a 32-bit word with a 28-bit heap address.

Word	Tag	Meaning
Process word	`ZERO`	the process `0`
	`BOOL` b	the Boolean value b
	`PTR` a	the process stored at heap address a
Name word	`QZERO`	the name `@0`
	`QBOOL` b	the name @b
	`QPTR` a	the name @(node at a)
	`VAR` i	the variable with de Bruijn index i (code only)

Closed Boolean names are immediates and never touch the heap.

Node kind	Fields
`PAR`	head: process word; tail: process word (a `PAR` or the last component)
`SEND`	chan: name word; payload: process word
`RECV`	chan: name word; pat: pattern-program address; body: process word; nvars; persistent bit
`DROP`	name: name word (always a `VAR`, by normalisation)
`BFN`	arity $k \le K$; support: k variable indices, ascending; table: 2^K bits

Every node carries a header with its kind, a *free-variable depth*, and a *sort* bit (the node is of sort Bool). The depth of a node is the least k such that every free variable index in it is below k; a node is *closed* when its depth is zero. Both fields are computed at interning time from the children: a variable i has depth $i + 1$; a receive with n pattern variables has depth $\max(d_{chan}, d_{body} - n, 0)$; every other node takes the maximum over its fields. A single closed

bit would not do: whether a receive is closed depends on whether its body's free variables are all bound by its own pattern, which a bit on the body cannot tell. The exact packing into block-RAM words is left to the implementation; it must be fixed in one shared definition used by the compiler, the reference model and the hardware.

Data. Section 5 adds words and nodes.

- **Immediates.** A word tag for integers that fit the payload of a word, in both the process and the name form. An integer that fits is always immediate and one that does not is always boxed, so the representation is unique.
- **Leaves.** `IBOX` (a 64-bit integer), `FIX` (a value and its format), and `BYTES` (a length and a sorted-by-position list of full chunks ending in one partial chunk, which is a unique chunking).
- **Constructors.** `TUP` (arity and fields), `CONS` (head, tail) with an immediate for the empty list, `SET` and `MAP` (sorted cons lists, without duplicates and with one entry per key).
- **Applications.** `APP` (operation and arguments), present only in open code.

The sort bit of the header becomes a sort field of four bits.

Class. The header also carries the node's class (Section 6.2), c bits wide. Addresses are local to a tile: a `PTR` or `QPTR` word means nothing in another tile, which is why terms cross between tiles as streams and not as words. Static addresses are the exception, being the same everywhere.

6.5 Heap, interner and reference counts

- **Heap.** An array of nodes in block RAM with a free list.
- **Interner.** `intern(node)` returns the address of the unique stored node equal to its argument, allocating if there is none. It is a multi-cycle operation: hash the node contents, probe a hash table in block RAM, and confirm any hit by a full field comparison. Equality is exact; the hash only selects the bucket, so there is no collision risk.
- **Normalising constructors.** The interner is the only way to create a node, and each constructor normalises: `PAR` performs the sorted insertion of Section 4; `BFN` minimises the support and collapses constants and projections; `DROP` of a closed name returns the named process.
- **Reference counts.** Terms are acyclic, so reference counting reclaims everything. A count is held per node in a separate RAM. Decrements that reach zero push the node on a deferred-free stack that is drained by a background rule, which bounds the work per cycle.

Each tile has its own heap, interner and counts. The static segment (Section 6.3) occupies the same addresses in all of them.

6.6 Tables

Each tile has two tables, each with a valid bit and a busy bit per entry.

- **Send table** $S[0..N_S)$: channel (closed name word), payload (closed process word).
- **Receive table** $R[0..N_R)$: channel (closed name word), pattern-program address, body (process word, code), environment address, persistent bit.

Every channel in a tile's tables is owned by that tile.

A receive in the soup is a *closure*: its body is shared code and its free variables are supplied by an environment, a heap-allocated vector of closed name words. Anything that becomes a *name* is fully instantiated instead (Section 6.11), since names must be canonical to be compared.

Lookup by channel is a comparison of one name word against all entries of the other table. At the table sizes of a tile this is a register array with N_R comparators.

Top-level processes that are Boolean values are inert. Each tile counts the ones that come to rest in it in a two-counter *residue* register, so that they remain observable.

6.7 Evaluator and service units

Evaluator. Each tile has an evaluator beside its instantiator. When instantiation closes an `APP` node, the evaluator replaces it by its value before the node is interned, so no closed application is ever stored.

- Additions, subtractions, comparisons, bitwise operations and fixed-point additions are combinational.
- Multiplications use the device's hard multipliers and take a fixed small number of cycles.
- Divisions, and every operation that walks a collection or a byte string, take a number of cycles bounded by the operand's size.

The evaluator works on jobs, so a long evaluation delays one continuation and nothing else: arrival and commitment continue around it.

Collections. The normalising constructors of sets and maps are the sorted-list constructor of parallel composition with one change each: a set drops duplicates, and a map keeps the later of two entries with one key. Collections longer than the parameter L raise a resource fault.

Service units. A service unit occupies one or more positions of the array and has a tile's router, inbox and outbox, so the mesh does not distinguish it. In place of tables and a matcher it has a function core. It takes a request packet, computes, and sends the reply to the owner of the return name. It keeps no state between requests and makes no choice, so it writes no decision records, and a request in service counts as a job.

6.8 Matcher and mask solver

The compiler turns each pattern into a *pattern program*, a short sequence of operations executed against a name word with an explicit cursor stack.

Operation	Effect
`NBIND`	bind the current name word to the next structural variable
`NEQ` i	fail unless the current name word equals structural variable i
`QZERO`	fail unless the current name word is `QZERO`
`QSEND`	fail unless the current name quotes a `SEND`; push the quoted payload, then continue with the channel
`BLEAF` T	raise `FAULT_SORT` unless the current name word is `QBOOL` b; set $mask \leftarrow mask \wedge (b\ ?\ T : \neg T)$
`END`	succeed if the cursor stack is empty

All Boolean leaves of one pattern are expressed by the compiler over a single ordered space of that pattern's mask variables, so their masks conjoin directly. The mask starts as all ones. The result of a successful run is the vector of structural bindings and the mask.

- The mask is the solution set: bit r is set exactly when the assignment $v_i \mapsto$ bit i of r solves every Boolean leaf.
- The pair is a redex if and only if the run succeeds and the mask is nonzero.
- The solver returns $\mathit{count} = \mathrm{popcount}(\mathit{mask})$ and, given $u < \mathit{count}$ from the arbiter, the index r of the u-th set bit. An occurrence of a mask variable outside any operator is a leaf whose table is a projection, and a non-linear Boolean pattern such as $v \,\&\&\, {\sim}v$ needs no special case. A variable that occurs under no operator is not a mask variable and is bound by `NBIND`, as in the AND gate of Appendix A.

The pure parts (mask update, popcount, select) are combinational functions and are among the Clash components of Section 7.

6.9 Arbiter and entropy

All nondeterminism of the calculus passes through one module per tile, with one operation: `choose(n)` returns $u \in [0, n)$ for $n \geq 1$. A tile uses it three times in a cycle: to pick a send, to pick a receive among those on the same channel, and to pick a solution row.

- **No pruning.** For every n and every $u < n$ there must be an entropy value for which `choose(n)` returns u. A priority encoder is forbidden: it would make the machine a refinement of the calculus and not bisimilar to it.
- **Entropy.** Each tile has three 32-bit maximal-length Galois LFSRs with taps 32, 22, 2 and 1, one per choice, with distinct nonzero seeds that also differ from tile to tile. A choice among n takes the register's value w and returns $\lfloor w\, n / 2^{32} \rfloor$. Each register advances 32 positions per clock, which is one fixed layer of exclusive-ors, so that successive cycles use disjoint bits. A hardware random source may replace a register behind the same interface.
- **Fairness.** Only possibilistic fairness is required (**D14**). The three-stage choice does not induce the uniform distribution over redexes. When grading is added, each stage takes a weight vector in place of a count, and weighting a pair by its solution count makes the composite uniform; the `choose` interface is defined so that this is a change of argument type only.
- **Weighted choice.** For grades (Section 5.4) the arbiter has a second operation, a weighted choice over a vector of weights. For the tropical algebra it chooses uniformly among the entries of least weight. For $\mathbb{R}_{\geq 0}$ it forms the exact sum W in a widened accumulator, takes $r = \lfloor w\, W / 2^{32} \rfloor$ from a sample w, and returns the entry in whose interval of the running sum r falls. An entry of weight zero is never chosen and every other entry may be. The no-pruning rule applies to entries of nonzero weight.
- **Replay.** In replay mode the arbiter ignores entropy and takes its results from the tile's decision log.

6.10 Concurrent reduction in a tile

A tile is not a state machine that steps through phases. It is a set of rules, each guarded by the readiness of what it reads and writes, and all of them fire in every cycle in which their guards hold. They communicate through the tables and three queues: inbox, job queue and outbox.

Rule	Action
`ARRIVE`	Take flits from the inbox and intern them. When a packet is complete, insert its send or receive into the tables.
`COMMIT`	The COMM rule. Remove one send and one receive that form a redex, write the decision record, and queue the continuation as a job.
`EXPLODE`	Take a job apart. Deliver each component to its owner: into the local tables, or into the outbox as a packet.
`DEPART`	Move flits from the outbox to the router.
`ROUTE`	Forward flits between the four links, the inbox and the outbox.
`SWEEP`	Reclaim nodes whose reference count is zero, when the interner is free.
`SCAN`	Search the tables for any redex, for the quiescence report only.

COMMIT. In each cycle:

1. Choose uniformly among the valid sends that are not busy.
2. Compare its channel with every valid receive that is not busy, and choose uniformly among those that match.
3. Run the pattern program of that receive on the quoted payload.
4. If the run succeeds with a nonzero mask, choose a solution row. In the same cycle: remove the send; remove the receive unless it is persistent; write the decision record; and queue the job (body, environment). The environment is that of the receive, extended by the structural bindings and, for each mask variable, the `QBOOL` of its bit in the row. Variables enter in a fixed order: structural variables in order of first occurrence in the pattern program, then mask variables in order of mask position, so that the last mask variable is the most recently bound.
5. Otherwise nothing happens in this cycle. The next cycle chooses afresh.

A tile commits at most one COMM per cycle (**D19**). A redex that rests in the tile has a nonzero probability of being the one chosen in every cycle in which the job queue has room, which is the no-pruning requirement.

A pattern program that reads only immediates and the node its cursor starts on is evaluated combinationally, and the whole of COMMIT is one cycle. A deeper program is run by the matcher over several cycles. The send and the receive under examination are marked busy for the duration and are not offered to another match; the removal, the record and the job are still written in one cycle, when the match ends.

EXPLODE. A job is a body and an environment, and stands for the instantiated body. A work stack holds (process word, environment) pairs.

- `PAR`: push head and tail.
- `SEND`: resolve the channel through the environment, instantiate the payload (Section 6.11), and find the owner of the channel. If this tile owns it, insert into S; otherwise write the send to the outbox as a packet.
- `RECV`: resolve the channel and find its owner. If this tile owns it, insert a closure into R. Otherwise instantiate the receive and write it to the outbox.
- `DROP` of variable i: look up the name. `QZERO`: nothing. `QBOOL`: increment the residue. `QPTR`: push the named process with an empty environment. This is the general drop that the replication encoding needs; because the quoted term is closed and shared, it costs no copying.

- `BFN`: evaluate the table at the environment values; increment the residue.
- `BOOL`: increment the residue. `ZERO`: nothing.

A job is exploded over several cycles while the tile goes on arriving and committing.

Shared resources. `ARRIVE` and `EXPLODE` both use the interner and both insert into the tables. Where they conflict `ARRIVE` goes first, as **D23** requires. The order is not observable, because neither rule chooses a redex: each only moves a component from one place where it already counts as part of the process to another.

What a cycle means. Across the array, the `COMMIT`s that fire in one cycle are pairwise independent redexes, one per tile at most. The effect of the cycle on the process the array stands for is their joint reduction, in any order. Every other rule leaves that process unchanged.

Quiescence. A tile is *idle* when its job queue, inbox and outbox are empty and `SCAN` has completed a full pass over its (send, receive) pairs, finding no redex, since its tables last changed. The array is quiescent when every tile is idle and every link is empty; this is a registered conjunction over the array. `SCAN` is sequential, but it only reports. It never chooses.

6.11 Instantiation

`inst(code, env)` returns the canonical closed term obtained by substituting the environment into the code. It is a post-order walk with an explicit stack that rebuilds nodes through the interner, tracking the number k of binders passed.

- A subterm whose free-variable depth is at most k is returned unchanged: it mentions no variable beyond the binders already passed. This is the common case and makes instantiation proportional to the open part of the code only.
- `VAR` i with $i < k$ is local and unchanged; with $i \geq k$ it is replaced by environment entry $i - k$, where entries are numbered from the most recently bound. Environment entries are closed, so no index shifting is needed.
- A `BFN` whose support becomes partly known is cofactored on the known variables and re-normalised; when fully known it becomes a `BOOL`.

6.12 Boundary

The environment is modelled as processes on channels that the namespace map assigns to *ports*.

- **Egress.** A send or receive whose channel is owned by a port is routed to the edge of the mesh like any other packet and leaves on that port's valid/ready interface.
- **Ingress.** A port injects a packet at the edge tile it is attached to, addressed to the owner of its channel. In the first stages the payload is restricted to immediates, and the sort tag is checked against the port's declared sort; a mismatch is dropped and counted.

6.13 Decision log and faults

Per tile. Each tile writes one decision record per `COMMIT`: cycle number, send slot, receive slot, solution row. Further streams report every insert into S or R with its slot, every residue increment, and every packet that departs or arrives, with a packet identifier. Together these let an external checker reconstruct the soup without reading the machine's internals.

The array's log (D24). The array's log is the union of the tiles' logs. Records of the same cycle are unordered, and rightly so: they are independent. The log is therefore a partial order of reductions and not a sequence.

Replay. Given the outputs of its arbiters, the array is a deterministic synchronous circuit. Replaying each tile's log, keyed on the cycle number, reproduces every cycle of the run.

Faults are sticky status bits. A fault in any tile halts every tile on the following cycle.

Fault	Raised when
`FAULT_SORT`	a Boolean operand or Boolean leaf meets a non-Boolean name. Unreachable for well-sorted programs (**D10**).
`FAULT_HEAP`	the heap, hash table or environment space is exhausted
`FAULT_STORE`	a send or receive table is full
`FAULT_STACK`	a work stack overflows
`FAULT_DATA`	a partial operation has no result (**D30**)

Resource faults are the only way the machine departs from the calculus, and they are detected, never silent.

Remark 2 (The sort flag is not a sort check)*.* A Boolean law can erase an ill-sorted operand before anything inspects it: $*v$ `xor` $*v$ normalises to `false` and $*v$ && `true` to $*v$, whatever v is later bound to. `FAULT_SORT` therefore fires only for operands that survive normalisation. Its never firing does not show that a program is well sorted, and Obligation 8 cannot be turned round. The sort checker carries the whole obligation.

6.14 Parameters

Parameter	Stage 1 value	Meaning
$W \times H$	2×2, then 10×10	array dimensions
P	4	ports
c	6	class width; at most 64 classes
K	6	maximum Boolean support and pattern mask variables
N_S, N_R	32, 32	send and receive table sizes, per tile
A	16 (later)	heap address width, per tile; stage 1 has no dynamic heap
E	16	maximum environment length
L	256	maximum length of a collection or of a byte string in chunks
D	16	work-stack depth
job queue	4	pending jobs per tile
inbox, outbox	16, 16	flits
flit	36 bits	one word and its framing
LFSR	3×32 bits	entropy state, per tile

7 Tooling and code organisation

7.1 Languages (D9)

Reference library (plain Haskell). Abstract syntax, normal forms, match sets, the reduction relation returning *all* redexes, the class automaton, and a model of the tile array. This is the executable semantics. The compiler and the checker are built on it.

Bluespec (BH). The whole of the hardware. The pure datapath (truth tables, masks, node packing, class tables) is written as BH functions in packages that hold no state, so that it

can be tested on its own and later mirrored in Clash. Everything with state is a module of rules. The calculus is a rewrite system, and Bluespec's guarded atomic rules express the rules of Section 6.10 directly; its scheduler derives which of them may fire together.

7.2 Rules for the Bluespec code

1. Every semantically visible choice goes through the arbiter module. The Bluespec scheduler resolves rule conflicts by a fixed urgency order, which is a priority encoder; it must never be the thing that decides which redex fires.
2. Every rule is marked in a comment as *reducing* (it changes the process the array stands for, by one redex) or *administrative* (it does not). Only `mkCommit` contains reducing rules.
3. Any urgency annotation must carry a comment stating why the conflict it resolves is administrative.
4. No rule on the inbound path may have a guard that depends on outbound state (**D23**).
5. The tile is one module, instantiated $W \times H$ times with its coordinates and seeds as its only parameters. There is no tile-specific logic and no logic outside tiles other than the port adapters and the fault and quiescence trees.
6. Each rule that changes the tables or the heap states, as an assertion, the invariants of Section 9.4 it relies on.
7. No vendor primitives. Memories are inferred. One clock, synchronous reset.

7.3 Suggested module list

Package or module	Kind	Responsibility
`TruthTable`	pure	table algebra and normalisation
`MaskSolve`	pure	mask, popcount, select
`NodeCodec`	pure	packing, hash, free-variable depth and sort bit
`ClassTables`	pure	the class automaton and the owner table
`DataOps`	pure	integer, fixed-point and byte operations
`mkHeap`	rules	node RAM, static segment, free list, reference counts
`mkInterner`	rules	hash-cons table, normalising constructors, classes
`mkTables`	rules	send and receive tables, channel lookup, residue
`mkMatcher`	rules	pattern-program evaluation
`mkCommit`	rules	the COMM rule
`mkExploder`	rules	instantiate, explode, deliver
`mkEvaluator`	rules	inline operations and collection walks
`mkService`	rules	the shell of a service unit, around a function core
`mkPacketiser`	rules	term streams out and in
`mkRouter`	rules	dimension-ordered wormhole routing
`mkArbiter`	rules	`choose`, LFSRs, replay input
`mkLog`	rules	decision, insert, residue and packet streams
`mkTile`	rules	one tile
`mkArray`	rules	the mesh, ports, fault and quiescence trees

7.4 Flow

Source text → compiler → memory image. Bluespec → Verilog → simulation (Icarus Verilog, Verilator) and synthesis (vendor or open flow). The same Verilog is the input to bounded formal

checks. The toolchain this draft was checked against is the Bluespec compiler 2026.07.1 with Icarus Verilog 13.

8 Compiler and image

The compiler is software and is in the trusted base for sorts only (**D10**); every other property of its output is checked dynamically by the verification harness.

1. Parse; resolve variables to de Bruijn indices.
2. Run the sort checker; record each pattern variable's sort.
3. Normalise (Section 4); reject Boolean supports above K.
4. Compile patterns to pattern programs. Assign mask variables their positions in the pattern's mask space and extend each leaf table to that space.
5. Intern all code into a heap image using the same interner algorithm as the hardware, so that addresses in the image are canonical.
6. Explode the top-level process into initial send and receive tables.
7. Emit block-RAM initialisation files for the heap, hash table, reference counts, pattern store and the two tables, plus a symbol file mapping source names to name words for the test harness.

Data. The compiler lowers f1r3lang's ground types, collections and method calls to the values and operations of Section 5. It evaluates closed applications itself, with the same functions as the hardware, converts literals to their representations (a Viterbi weight to its negated logarithm, a decimal to its `Fix` format), and turns each use of a system contract into a send on the corresponding service channel.

Namespaces and placement. Three further steps serve the array.

1. Compile the namespace map to the class tables and the owner table, and check that it is a partition (Section 6.2).
2. Assign numerals to symbols. The default is order of first occurrence. The placing option chooses them to keep channels that exchange messages in the same tile or in neighbouring ones, since under the default map the numeral decides the tile.
3. Emit the static segment once, for every tile, and for each tile its initial tables: every initial component of the program is placed in the tile that owns its channel.

9 Verification

9.1 What "verified" means

Let s be the state of the array at a clock edge. The process it stands for is

$$\alpha(s) \;=\; \prod_{t}\Big(\prod_{i\in S_t} c_i!(q_i) \;\mid\; \prod_{j\in R_t} \mathsf{inst}(recv_j, env_j) \;\mid\; \prod_{k\in J_t} \mathsf{inst}(body_k, env_k) \;\mid\; \texttt{true}^{\,t_t} \mid \texttt{false}^{\,f_t}\Big) \;\mid\; \prod_{p} term(p)$$

where t ranges over tiles, J_t is the part of tile t's jobs not yet delivered, (t_t, f_t) is its residue, and p ranges over the packets between an exploder and a table, each standing for its send or receive however much of it has been transmitted. Every clock edge is a boundary (**D24**).

Obligation 3 (Soundness)**.** If the array goes from clock edge s to the next edge s', then $\alpha(s')$ is $\alpha(s)$ reduced by exactly the redexes whose `COMMIT`s fired in that cycle, and these are pairwise independent. If none fired, $\alpha(s') \equiv \alpha(s)$.

Obligation 4 (No pruning)**.** If $\alpha(s) \to Q$ by a redex whose send and receive rest in the tables of a tile whose job queue has room, that tile commits that redex in the next cycle with nonzero probability.

Obligation 5 (Delivery)**.** Absent faults, every component of every job and every packet comes to rest in the tables or residue of its owner within a bounded number of cycles. No packet is lost, duplicated or held for ever.

Obligation 6 (Quiescence)**.** The quiescent bit is set at a clock edge if and only if $\alpha(s)$ has no redex.

Obligation 7 (Replay)**.** Two runs from the same image with the same ingress sequence and the same per-tile decision logs pass through identical states at every clock edge.

Obligation 8 (Sort safety)**.** For a well-sorted image, `FAULT_SORT` is unreachable.

Obligation 9 (Placement independence)**.** For a fixed program, the processes reachable as $\alpha(s)$, up to $\equiv$, do not depend on the dimensions of the array, on the namespace map, or on the assignment of numerals to symbols.

Obligation 10 (Data)**.** Evaluation is a function. A closed application has one value, the same in every tile, in every service unit, in the compiler and in the reference; and two values of an inline type are equal exactly when their words are.

Obligations 3, 4 and 5 together say that the fault-free array, observed at clock edges, is bisimilar to the calculus: the administrative moves are silent, and every redex of the calculus is, after a bounded delay, a redex some tile may commit. Obligation 9 says the tiling is an implementation matter and not a change of language. Obligations 1 and 2 are the supporting facts about the language itself.

9.2 Layers of evidence

Layer	Method	Discharges
V1	Property tests of the reference normaliser against a brute-force decision of $\equiv$ on small terms; exhaustive for Boolean code with $k \leq 3$	Ob. 2 (reference)
V2	The Clash functions are the reference functions: test in GHC, then check generated Verilog against them by random co-simulation	datapath correctness
V3	Interner test: intern random raw terms through the machine's normalising constructors alone; addresses equal iff reference normal forms equal, and readback equals the reference normal form	Ob. 2 (hardware)
V4	Step-refinement checker (below) on random and directed programs	Ob. 3, 6
V5	Formal cover on the arbiter: for every $n \leq 64$ and $u < n$ a witness entropy value exists	Ob. 4 (arbiter)
V6	Redex coverage: on small programs, every reference redex at every visited state is taken under some seed	Ob. 4 (system)
V7	Replay test: record, replay, compare boundary traces	Ob. 7
V8	Assertions for the invariants below, in simulation and bounded model checking	supports all
V9	Bounded bisimulation check, in the reference, of persistent receive against the encoding; paper proof to follow	Ob. 1

Layer	Method	Discharges
V10	Instantiation commutes with normalisation, in the reference and in the machine, against textbook substitution on raw terms	Ob. 3 (instantiation)
V11	Mutation: each of a set of bugs injected into the machine and into the reference must fail some layer	adequacy of V1–V10
V12	Placement independence: the same programs on 1×1, 2×2 and 10×10 arrays and under different maps reach the reference transition graph, administrative moves aside	Ob. 9
V13	Mesh: random traffic with every packet accounted for; bounded model checking of deadlock freedom on small meshes	Ob. 5
V14	Concurrency: runs in which several tiles commit in one cycle are produced and accepted, and commits per cycle are measured	**D20**
V15	Evaluator: every operation against the reference on random and boundary operands; exhaustively for an 8-bit instantiation of `Int` and `Fix`	Ob. 10
V16	Collections: a set or map built from the same elements in any order interns to the same word	Ob. 10, 2
V17	Service units: published test vectors for each hash and signature scheme; random agreement with a software implementation	Ob. 10
V18	Weighted choice: sums independent of candidate order; no entry of nonzero weight is unreachable; frequencies track weights	Ob. 4

9.3 The step-refinement checker

The checker is a testbench component built on the reference library. It reads only the streams of Section 6.13, and the heap contents needed to decode the addresses they report.

1. It maintains the abstract state: a map from each tile's table slots to reference terms, the residue counts, the undelivered part of each job, and the packets in flight. This is α computed incrementally.

2. In each cycle it collects the decision records of all tiles. For each it checks that the triple is a redex of the abstract state: channels equivalent, the row in the reference match set. Records of one cycle come from different tiles and so use different sends and receives.

3. It applies those redexes together, turning each into a job that stands for the reference reduct.

4. It checks every administrative event for conservation: each component a job delivers is a component of that job's reduct, and is consumed from it; each packet that arrives equals one that departed, and is consumed from the flight set; nothing enters a table by any other way.

5. When the array reports quiescence it checks that the reference state has no redex.

Any discrepancy is a failure with the offending cycle and tile, which the replay logs reproduce exactly.

9.4 Invariants

I1. Every stored node is in normal form.

I2. No two distinct heap addresses hold equal nodes.

I3. Each reference count equals the number of references from nodes, tables, environments and stacks.

I4. Every channel word and every payload word in the store is closed.

I5. A `DROP` node's name is a `VAR`; a `BFN` has minimal, ascending support, is not constant and is not a projection.

I6. A persistent receive is never removed by a step.

I7. An entry marked busy is under examination by exactly one match in progress.

I8. The mask is nonzero at COMMIT, and the chosen row is a set bit.

I9. Exactly one decision record is written per `COMMIT`.

I10. Every channel in a tile's tables is owned by that tile.

I11. The static segment is identical in every tile and is never freed.

I12. A packet that departs arrives exactly once, at its destination. Flits of different packets do not interleave on a link.

I13. A tile writes at most one decision record per cycle.

I14. A tile's inbound path is never stalled while its heap and tables have room.

9.5 Directed test programs

Program	Checks
Forwarder, AND, XOR gates	forward evaluation; one solution per match
Half adder, full adder	composition of gates through channels
Inverse XOR (Appendix A)	two solutions, both reached; empirical frequencies
Inverse AND on `false`	three solutions, all reached
Unsatisfiable pattern v && $\sim v$ on `true`	no redex; send remains; quiescence
Two receivers, one send	COMM race; both outcomes reached
Replication by encoding	general drop; an unbounded run. Hash-consing shares every copy, so the heap does not grow: the run ends on the receive-table bound when the replicated process leaves receives behind, and otherwise never
Persistent receive	same traces as the encoding, fewer steps
Equal names, different syntax	@(`true` && `false`) and @`false` as channels are the same channel
Ill-sorted image (negative test)	`FAULT_SORT` is raised and halts
Forwarder across two tiles	a send and its receive start in different tiles; one packet; one COMM
Ping-pong between two tiles	sustained traffic in both directions; no deadlock; packet accounting
Independent gates in different tiles	commits in the same cycle; a partial-order log
One program on three array sizes	the same reference graph on each
A map that is not a partition (negative test)	rejected by the compiler
Integer adder and comparator	inline evaluation; the result of one COMM
Counter with a persistent receive	`Int` state carried by messages
Division by zero (negative test)	`FAULT_DATA` is raised and halts
A set built in two orders, used as a channel	equal names; one COMM
Hash request and reply across the mesh	a service unit as a channel

10 Staging

Stage	Content	Exit criterion
1. Boolean-core array	Names are symbols, Boolean immediates and immediate integers; shallow patterns; static code only, with no dynamic heap; fixed-size packets. All of the concurrency: tiles, mesh, the default namespace map, per-tile logs, quiescence. Combinational `Int` operations. Arrays of 1×1, 2×2, then 10×10	V4 to V8, V12 to V15 pass on gate, inverse-gate and counter programs, in simulation and on an FPGA
2. Reflection and data	Per-tile heap, interner with classes, reference counts, instantiation, term-stream packets, general drop, general namespace maps. Boxed `Int`, `Fix`, `Bytes`, strings, tuples, lists; the full evaluator	V1, V3, V10 pass; replication by encoding runs across tiles
3. Full patterns and collections	`QSEND` and deeper patterns with the multi-cycle matcher; tuple and list patterns; sets and maps; ingress of quoted terms	V9, V16 pass; full directed suite
4. Service units	Hash and signature units as tiles; the vector unit	V17 passes
5. Grades	Weighted choice in the arbiter; the graded `where` clause	V18 passes
6. Typed fast path	Channels typed Boolean-only are elaborated to one-bit links and gates inside a tile; the tuple-space machine remains the fallback	elaborated and interpreted runs agree under the checker

Later attachments, not staged here: cost accounting as a debit at `COMMIT`, and rent on a namespace as a charge on what rests in its tile; the interfering resolver for complex grades; floating point as a service unit; the combinator instruction set as an alternative code representation in the same heap.

11 Open questions and recorded limits

Open: matching under a binder. The pattern `for`$(v \leftarrow p)\, d$ matches a receive and its body. The body of the matched receive has a free bound variable, so a variable of d cannot simply bind "the name of the body". The candidate readings are: bind the body as an abstraction; or allow d to mention v and match up to renaming. The machine supports neither until this is settled, and the pattern is rejected by the compiler in the meantime.

Open: the formula fragment. **D21** takes "structural formula" to mean the fragment of Section 6.2: no behavioural modality, no counting of components, and nothing to say about open

code. It was chosen because a tile can then decide ownership from a name alone, in one lookup. If namespaces in use need more than this, the class automaton is the thing that has to grow.

Open: skew. Nothing stops a map, or a program, from sending most of the traffic to one tile. A tile commits one COMM per cycle whatever its load, so a skewed map gives a sequential machine with a large idle border. The compiler's placing option addresses static programs only.

Open: which resolution algebras come first. The grade types of Section 5.4 cover six algebras in fixed-point. Which of them the first hardware must resolve decides the width of the arbiter's accumulator and whether the limited range of `Fix` is acceptable for $\mathbb{R}_{\geq 0}$, where rates may span many orders of magnitude.

Open: failure of a partial operation. **D30** halts the array on a division by zero. A node aborts only the deployment that erred. Doing the same here needs a notion of deployment in the machine, which arrives with cost accounting.

Recorded limits. These follow from confirmed decisions and are listed so that an implementer does not mistake them for oversights.

- **Board.** None is fixed (**D16**). Block-RAM budgets are set per board through the parameters.
- **Trust.** The compiler is trusted for sorts (**D10**). A self-defending chip would need a hardware loader and a type system checkable in time linear in the image.
- **Support bound.** $K = 6$ (**D12**). Lifting it needs reduced ordered decision diagrams in the interner.
- **Fairness.** Possibilistic only (**D14**). A prescribed distribution over redexes enters through the arbiter's weight interface.
- **Top-level Booleans.** Kept in the residue counters.
- **Operator patterns.** Closed Boolean values only (**D13**).

12 Status and changes

Status. A reference implementation, `rhoref` 0.1.0, exists: the executable semantics, a word-level model of *one* tile with a sequential step, and the step-refinement checker, in plain Haskell. It conforms to draft 0.3 and has no data beyond the Booleans. Its semantics, normal forms, matcher and instantiation carry over unchanged; its machine model and checker must be extended to the array, the class automaton and the per-cycle obligations of this draft before any hardware is compared with it. No Bluespec has been written. Nothing has been proved; Obligation 1 in particular is tested and not yet proved.

Draft 0.5: builtin data types.

1. **Aim.** The machine is the target of a compiler from f1r3lang, which is the kernel calculus with graded `where` clauses and data.
2. **A new section** (Section 5) gives the test a type must pass to be builtin, the two kinds of builtin, the types, the grade representations, the laws that enter congruence, and what patterns may do.

3. **The machine** gains data words and nodes (Section 6.4), an evaluator in every tile and service units as tiles (Section 6.7), weighted choice in the arbiter (Section 6.9), and `FAULT_DATA`.
4. **Namespaces** may class leaf data by a residue of its content (**D31**).
5. **Verification** gains Obligation 10, layers V15 to V18 and five directed programs. **Staging** grows to six stages.
6. **Floating point** is considered and deferred, with reasons (**D32**).

Draft 0.4: the concurrent tile array.

1. **Tiling** (**D17**, **D19**, Sections 6, 6.3). The single machine is replaced by an array of identical tiles on a mesh, about a hundred by default, each with about a hundredth of the memory.
2. **Namespaces** (**D18**, **D21**, Section 6.2). Each tile owns a namespace given by a structural formula. The formulae are compiled to a class automaton that runs at interning time, so ownership is a table lookup.
3. **Concurrency** (**D20**, **D24**, Section 6.10). The sequential step is gone. Tiles commit in the same cycle; inside a tile the stages are concurrent rules; every clock edge is a boundary; the log is a partial order. Parallel reduction is no longer a late stage but the first.
4. **Distribution** (**D22**, **D23**). Heaps are per tile, terms cross as streams, static code is replicated, routing is dimension-ordered, and the inbound path never waits on the outbound.
5. **Language** (**D9**). The whole design is in Bluespec; Clash is optional and later.
6. **Verification**. Obligations are restated per cycle; delivery and placement independence are new; layers V12 to V14, invariants I10 to I14 and five directed programs are added.
7. **Entropy** is three registers per tile advancing 32 positions per clock, in place of one register advanced per choice.

Draft 0.3: corrections. Each was found by building or testing the reference, which implements the corrected reading.

1. **Free-variable depth** (Sections 6.4, 6.11). The node header carries a free-variable depth in place of a closed bit, which cannot be computed for a receive from its children.
2. **The sort flag is incomplete** (Section 6.13, **D10**). Boolean laws can erase an ill-sorted operand, so `FAULT_SORT` is a debugging aid and not a sort check.
3. **Mask variables** (Sections 3.4, 6.8, 6.10). Draft 0.2 said a bare Boolean-sorted variable is a projection leaf, while its appendix bound one with `NBIND`. The rule is now stated once: a variable is a mask variable exactly when it occurs under a Boolean operator in the pattern. The order in which variables enter the environment is fixed.
4. **A Boolean leaf on a non-Boolean** (Sections 3.4, 6.8) is a sort fault, not a failed match.
5. **Replication by encoding** (Section 9) ends on the receive-table bound, or never; it does not exhaust the heap.
6. **Entropy sampling** (Section 6.9) is specified: the LFSR polynomial, the map from a sample to a choice, and the reference model's 32 positions per choice.

The verification plan gains two layers. V10 was added because instantiation is where normal forms are re-established at run time. V11 was added because two injected bugs survived the first suite, which showed that the interner had been tested only on terms the reference had already normalised; V3 is restated accordingly.

A Worked examples

Let x, y, z, l, r be distinct closed names. Both examples are shown within one tile. When the names fall in different namespaces the same reductions happen, with sends and receives travelling to their owners first.

An AND gate

$$\texttt{for}(a \leftarrow x)\,\texttt{for}(b \leftarrow y)\,z!(*a \mathbin{\&\&} *b) \;\mid\; x!(\texttt{true}) \;\mid\; y!(\texttt{false})$$

- **Code.** Inside the inner body, b has index 0 and a has index 1. The payload is a `BFN` with support $(0, 1)$ and table $T = 1000_2$: only row $r = 3$, where both are true, is set.
- **Initial soup.** One receive on x with pattern program `NBIND; END` and an empty environment; sends $(x, \texttt{BOOL}\ 1)$ and $(y, \texttt{BOOL}\ 0)$.
- **Step 1.** The send on y has no receive, so the only redex is on x. The mask is all ones over zero Boolean variables, one solution. The environment becomes [@`true`]. Explode inserts a receive closure on y with that environment.
- **Step 2.** COMM on y. The environment becomes [@`true`, @`false`], so index 0 (b) is false and index 1 (a) is true: row $r = 2$. Instantiating the payload reads bit 2 of T, which is 0. Explode inserts the send $(z, \texttt{BOOL}\ 0)$.
- **Result.** $z!(\texttt{false})$, and the machine is quiescent.

An inverse XOR

$$\texttt{for}(@(p\ \texttt{xor}\ q) \leftarrow x)\,(l!(*p) \mid r!(*q)) \;\mid\; x!(\texttt{true})$$

- **Pattern program.** `BLEAF` T; `END`, with p at mask position 0, q at position 1 and $T = 0110_2$.
- **Match.** The payload is `QBOOL` 1, so the mask is $T = 0110_2$: rows 1 and 2, count 2.
- **Choice.** The arbiter returns $u \in \{0, 1\}$. With $u = 0$ the row is 1: $p = \texttt{true}$, $q = \texttt{false}$. With $u = 1$ the row is 2: $p = \texttt{false}$, $q = \texttt{true}$.
- **Result.** Either $l!(\texttt{true}) \mid r!(\texttt{false})$ or $l!(\texttt{false}) \mid r!(\texttt{true})$. Both must be reachable, and the row is recorded in the log. Had the payload been `false`, the mask would be $\neg T = 1001_2$ and the two outcomes would be the equal pairs.

The second example is a one-bit entropy tap written in the calculus itself: the quality of the machine's random source is observable by a program.